

MTH-262 Statistics and Probability Theory

Lecture 13



Outline of this lecture

In this lecture we will cover the following topics:

- Conditional probability distribution
 - For discrete random variable
 - For Continuous random variable
- Statistical independence of random variables

Conditional Probability Distribution

We revisit the definition of conditional probability as stated in Chapter 2,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) > 0,$$

where A and B are now the events defined by $X = x$ and $Y = y$, respectively, then

$$P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0,$$

where X and Y are discrete random variables.

It is not difficult to show that the function $f(x, y)/g(x)$, which is strictly a function of y with x fixed, satisfies all the conditions of a probability distribution. This is also true when $f(x, y)$ and $g(x)$ are the joint density and marginal distribution, respectively, of continuous random variables.

Conditional Probability Distribution

it is extremely important that we make use of the special type of distribution of the form $f(x, y)/g(x)$ in order to be able to effectively compute conditional probabilities. This type of distribution is called a conditional probability distribution; the formal definition follows.

Let X and Y be two random variables, discrete or continuous. The **conditional distribution** of the random variable Y given that $X = x$ is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

Conditional Probability Distribution of discrete random variable (Example)

Question: Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

- (a) the joint probability function $f(x, y)$,
- (b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) \mid x + y \leq 1\}$.
- (c) find the conditional distribution of X , given that $Y = 1$, and use it to determine $P(X = 0 \mid Y = 1)$.

Note: Part (a) and (b), we already solved in one of the previous lecture.

Conditional Probability Distribution of discrete random variable (Example)

All corresponding probabilities of the joint distribution are given in the following table:

Joint Probability Distribution

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Conditional Probability Distribution of discrete random variable (Example)

Solution: We need to find $f(x|y)$, where $y = 1$. First, we find

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Now

$$f(x|1) = \frac{f(x, 1)}{h(1)} = \left(\frac{7}{3}\right) f(x, 1), \quad x = 0, 1, 2.$$

Therefore

$$f(0|1) = \left(\frac{7}{3}\right) f(0, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2}, \quad f(1|1) = \left(\frac{7}{3}\right) f(1, 1) = \left(\frac{7}{3}\right) \left(\frac{3}{14}\right) = \frac{1}{2},$$
$$f(2|1) = \left(\frac{7}{3}\right) f(2, 1) = \left(\frac{7}{3}\right) (0) = 0,$$

Conditional Probability Distribution of discrete random variable (Example)

.....and the conditional distribution of X , given that $Y = 1$, is:

x	0	1	2
$f(x 1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

Finally,

$$P(X = 0 | Y = 1) = f(0|1) = \frac{1}{2}.$$

Therefore, if it is known that 1 of the 2 pen refills selected is red, we have a probability equal to $1/2$ that the other refill is not blue.

Conditional Probability Distribution of continuous random variable (Example)

Question: The joint density for the random variables (X, Y) , where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, is

$$f(x, y) = \begin{cases} 10xy^2, & 0 < x < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal densities $g(x)$, $h(y)$, and the conditional density $f(y|x)$.
- (b) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

Conditional Probability Distribution of continuous random variable (Example)

Solution: (a) By definition

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy = \int_x^1 10xy^2 \, dy \\ &= \frac{10}{3} xy^3 \Big|_{y=x}^{y=1} = \frac{10}{3} x(1 - x^3), \quad 0 < x < 1, \end{aligned}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^y 10xy^2 \, dx = 5x^2 y^2 \Big|_{x=0}^{x=y} = 5y^4, \quad 0 < y < 1.$$

Now

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1 - x^3)} = \frac{3y^2}{1 - x^3}, \quad 0 < x < y < 1.$$

Conditional Probability Distribution of continuous random variable (Example)

(b) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 unit.

$$\begin{aligned} P\left(Y > \frac{1}{2} \mid X = 0.25\right) &= \int_{1/2}^1 f(y \mid x = 0.25) dy \\ &= \int_{1/2}^1 \frac{3y^2}{1 - 0.25^3} dy = \frac{8}{9} \end{aligned}$$

Statistical Independence

Let X and Y be two random variables, discrete or continuous, with joint probability distribution $f(x, y)$ and marginal distributions $g(x)$ and $h(y)$, respectively. The random variables X and Y are said to be statistically independent if and only if

$$f(x, y) = g(x) \cdot h(y)$$

for all (x, y) within their range.

Statistical Independence (Proof)

If $f(x|y)$ does not depend on y then $f(x|y) = g(x)$ and $f(x, y) = g(x)h(y)$. The proof follows by substituting:

$$f(x, y) = f(x|y) \cdot h(y)$$

into the marginal distribution of X . That is,

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_{-\infty}^{\infty} f(x|y)h(y) \, dy.$$

If $f(x|y)$ does not depend on y , we may write

$$g(x) = f(x|y) \int_{-\infty}^{\infty} h(y) \, dy.$$

since $h(y)$ is the probability density function of Y $\int_{-\infty}^{\infty} h(y) \, dy = 1$. Therefore,

$$g(x) = f(x|y) \text{ and then } f(x, y) = g(x)h(y).$$

Statistical Independence of discrete random variables

Reconsider the joint probability distribution of a previous example.

Joint Probability Distribution

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
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Statistical Independence of discrete random variables

From Table (see previous slide) we find the three probabilities $f(0, 1)$, $g(0)$, and $h(1)$ to be:

$$f(0, 1) = \frac{3}{14},$$

$$g(0) = \sum_{y=0}^2 f(0, y) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$h(1) = \sum_{x=0}^2 f(x, 1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Clearly, $f(0, 1) \neq g(0)h(1)$, and therefore X and Y are not statistically independent.

Statistical Independence

Let $X_1, X_2, \dots, X_n$ be n random variables, discrete or continuous, with joint probability distribution $f(x_1, x_2, \dots, x_n)$ and marginal distribution $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$, respectively. The random variables $X_1, X_2, \dots, X_n$ are said to be mutually statistically independent if and only if

$$f(x_1, x_2, \dots, x_n) = f_1(x_1)f_2(x_2) \cdots f_n(x_n)$$

for all $(x_1, x_2, \dots, x_n)$ within their range.

Question 3.75

- A chemical system that results from a chemical reaction has two important components among others in a blend. The joint distribution describing the proportions X_1 and X_2 of these two components is given by

$$f(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Give the marginal distribution of X_1 .
- (b) Give the marginal distribution of X_2 .
- (c) What is the probability that component proportions produce the results $X_1 < 0.2$ and $X_2 > 0.5$?
- (d) Give the conditional distribution $f_{X_1/X_2}(x_1/x_2)$.

$$(a) \quad g(x_1) = \int_{x_1}^1 2 \, dx_2 = 2(1 - x_1), \text{ for } 0 < x_1 < 1.$$

$$(b) \quad h(x_2) = \int_0^{x_2} 2 \, dx_1 = 2x_2, \text{ for } 0 < x_2 < 1.$$

$$(c) \quad P(X_1 < 0.2, X_2 > 0.5) = \int_{0.5}^1 \int_0^{0.2} 2 \, dx_1 \, dx_2 = 2(1 - 0.5)(0.2 - 0) = 0.2.$$

$$(d) \quad f_{X_1|X_2}(x_1|x_2) = \frac{f(x_1, x_2)}{h(x_2)} = \frac{2}{2x_2} = \frac{1}{x_2}, \text{ for } 0 < x_1 < x_2.$$